

Data visualisation and exploration

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Introduction to Data Visualization & Data Sources

Session 1

What is data visualisation?

Data visualisation is the process of **representing data** and information through visual elements like **charts, graphs, and maps**. It transforms complex datasets into clear, accessible visuals, making it easier **to identify patterns, trends, and outliers**. This helps people **make faster, more informed decisions** across various fields.



Data to graph

DATA: BY THE NUMBERS

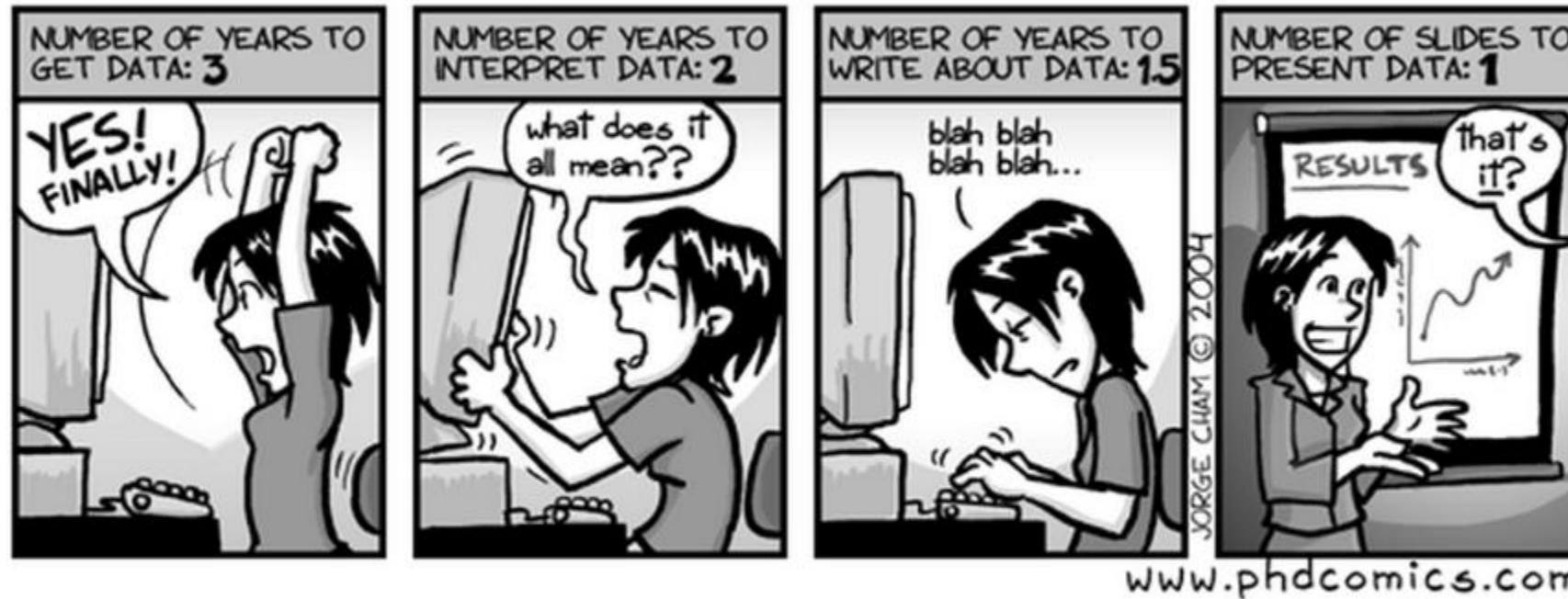


Figure 2.1: Data By the Numbers (by Jorge Cham)

Why data visualization is needed?

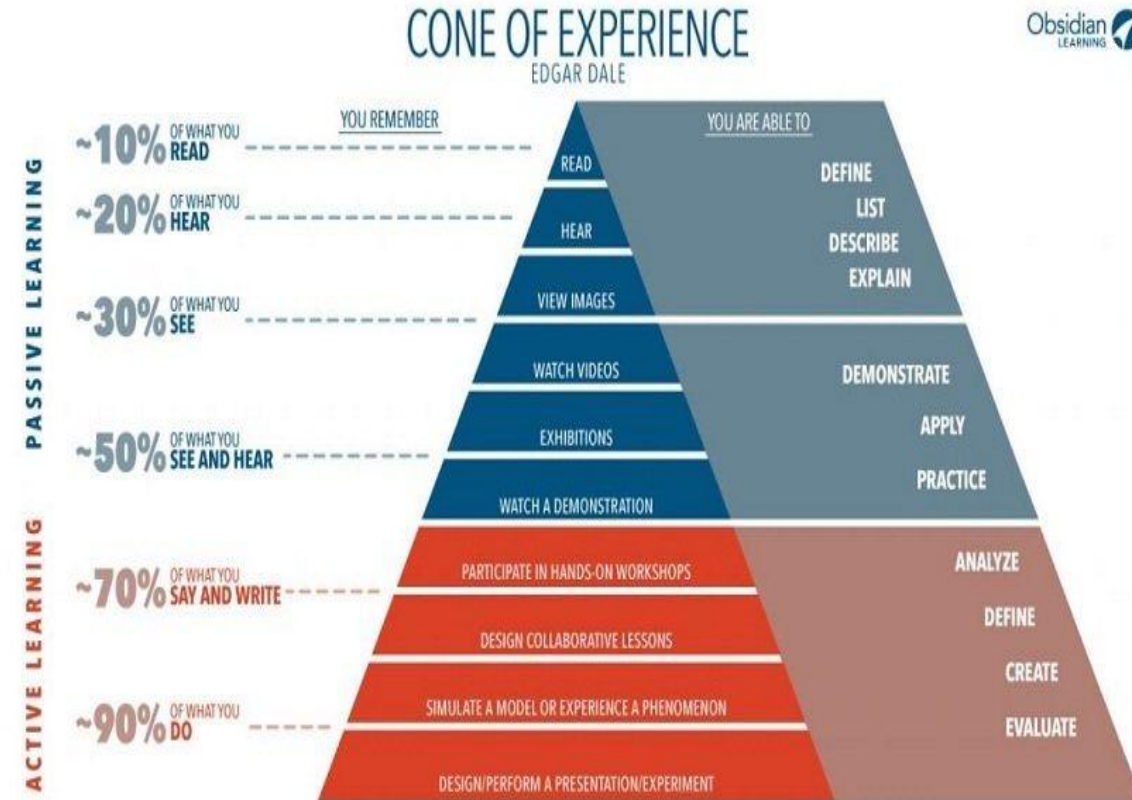
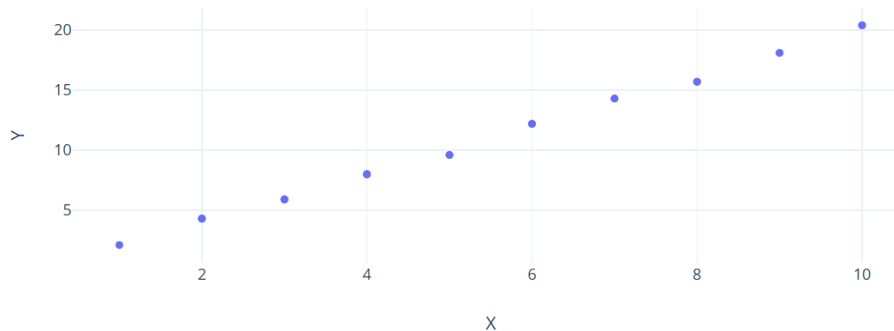
Our brains are wired for visuals:

- We process images **60,000× faster** than text

Trends, outliers, clusters, and gaps stand out instantly in a chart

Tables may hide insights that a simple plot can reveal

1	2.1
2	4.3
3	5.9
4	8.0
5	9.6
6	12.2
7	14.3
8	15.7
9	18.1

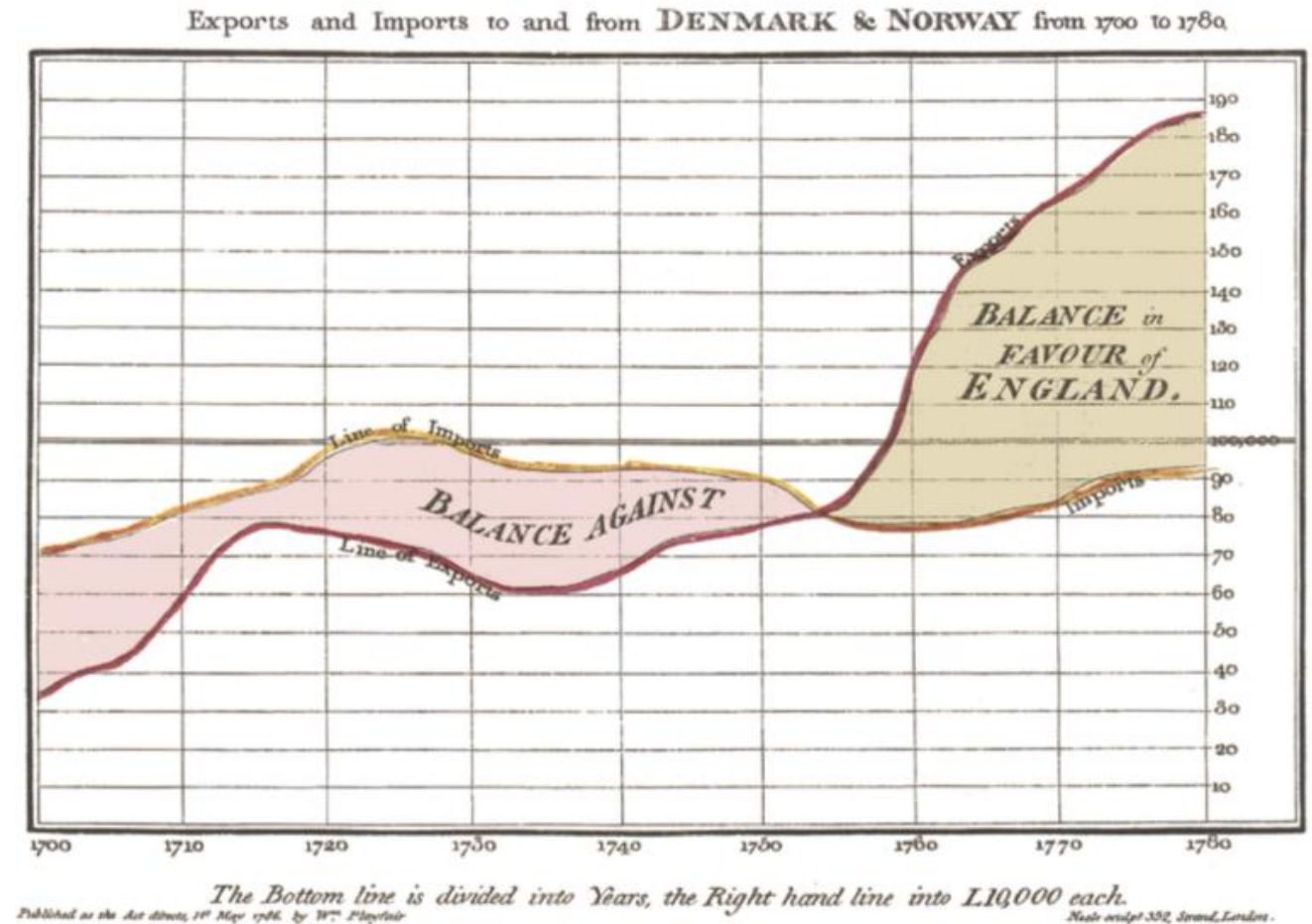


Is it something new?

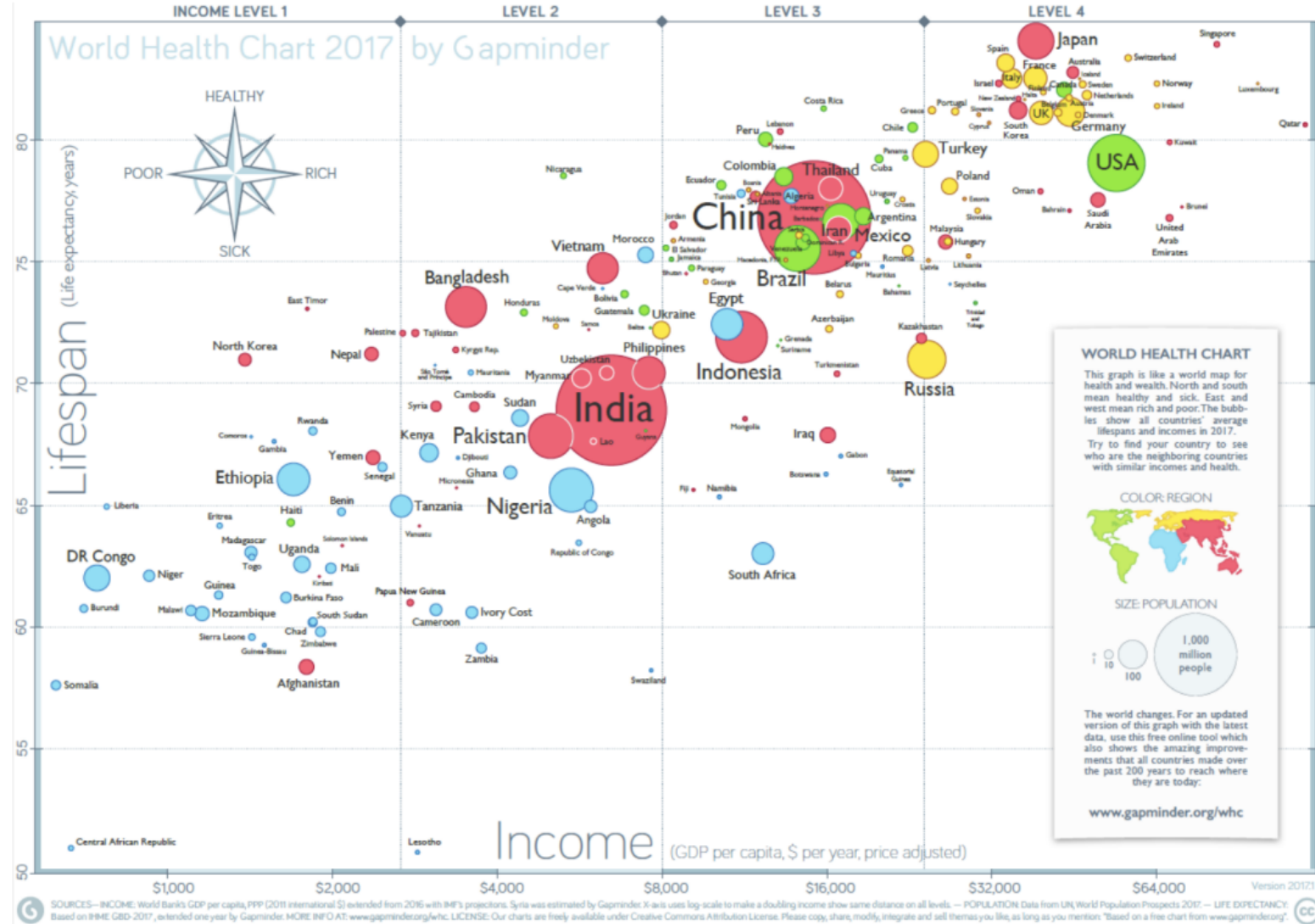
Given the recent explosion in data availability and visualization tools, one might assume that data visualization is a modern development.

However, this is not the case — data visualization has evolved over time, building on earlier methods to shape the tools and trends we use today.

For example, as early as the 18th century.



Now



Real world application

Real-World Applications of Data Visualization



Business Analytics



Retail & E-commerce Insights



Healthcare & Patient Tracking



Financial Data Visualization



Sports Performance Analytics



Government & Public Sector Analysis



Public Policy Track climate change trends and carbon emissions



Business Sales dashboards, customer behavior



Health Epidemic tracking, patient diagnostics, drug effectiveness



Finance Stock market trends, risk visualization, fraud detection



Human Rights Visualizing refugee flows or conflict zones



Engineering Sensor data monitoring, anomaly detection



Research Experimental results, network analysis



Journalism Data stories (e.g., elections, education access)

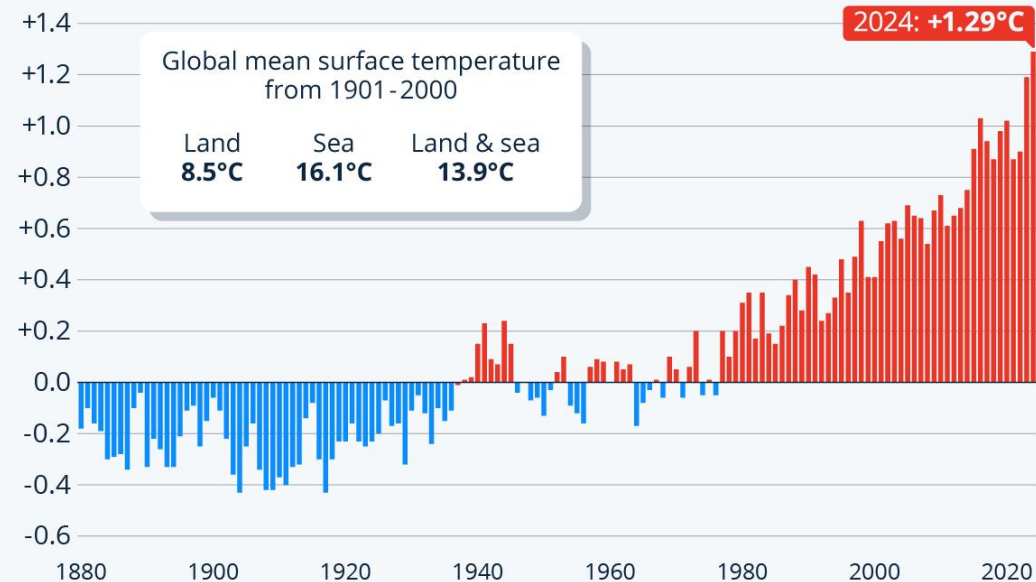
Importance of data visualisation

- **Simplifies Complex Data:** Converts large and complicated datasets into charts and graphs that are easier to understand.
- **Highlights Patterns and Trends:** Makes it easier to spot relationships and trends that are hard to see in raw numbers.
- **Saves Time:** Enables quick interpretation of data, reducing the need to scan through lengthy tables.
- **Enhances Communication:** Helps share insights clearly, even with people who aren't data experts.
- **Tells a Clear Story:** Guides the viewer through the data logically, supporting better understanding and decisions..

Importance of data visualisation

2024 Was the Planet's Warmest Year in Recorded History

Global land and ocean surface temperature anomalies
(in degrees Celsius compared to the 1901-2000 average)

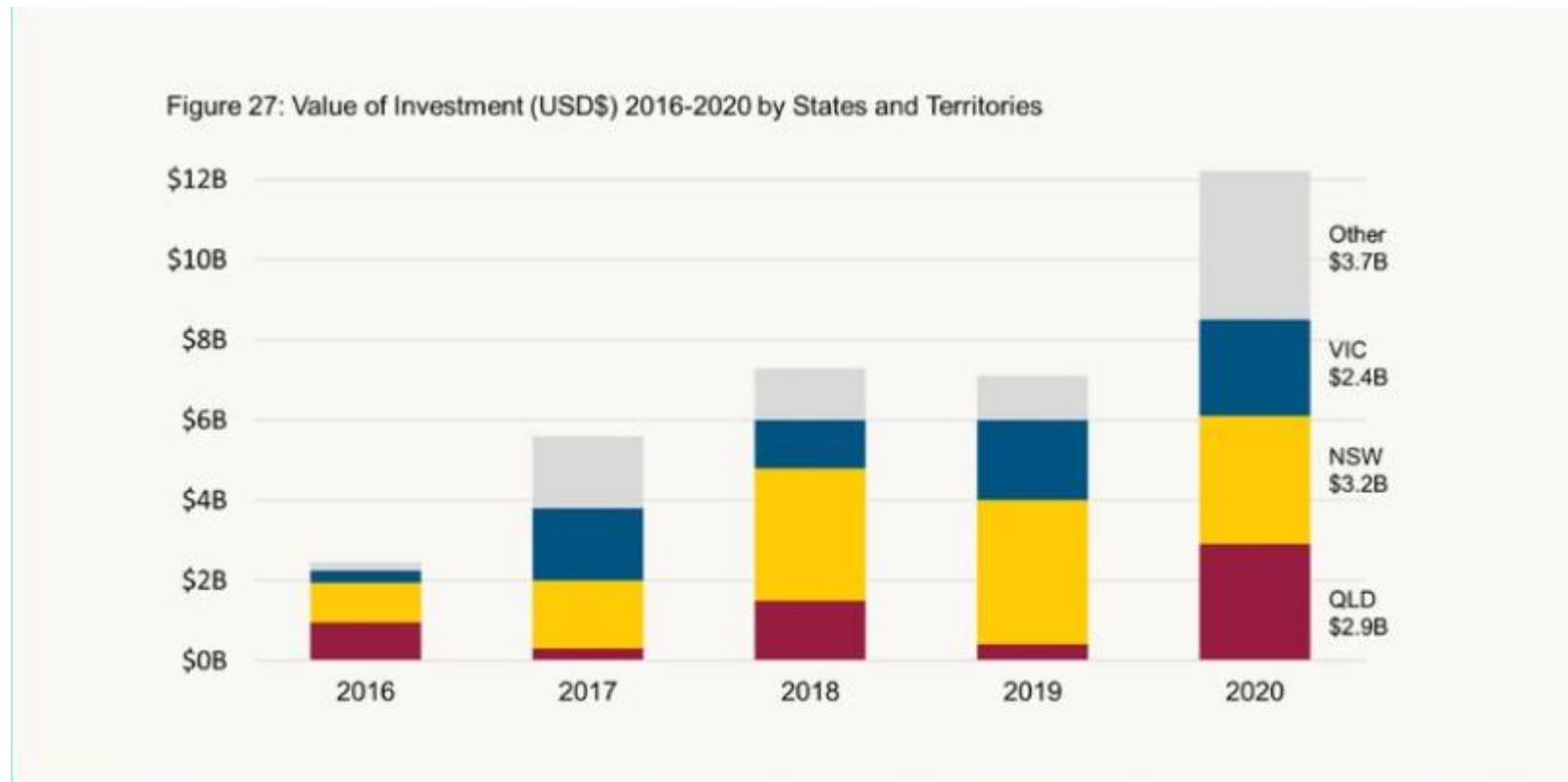


Source: NOAA National Centers for Environmental Information



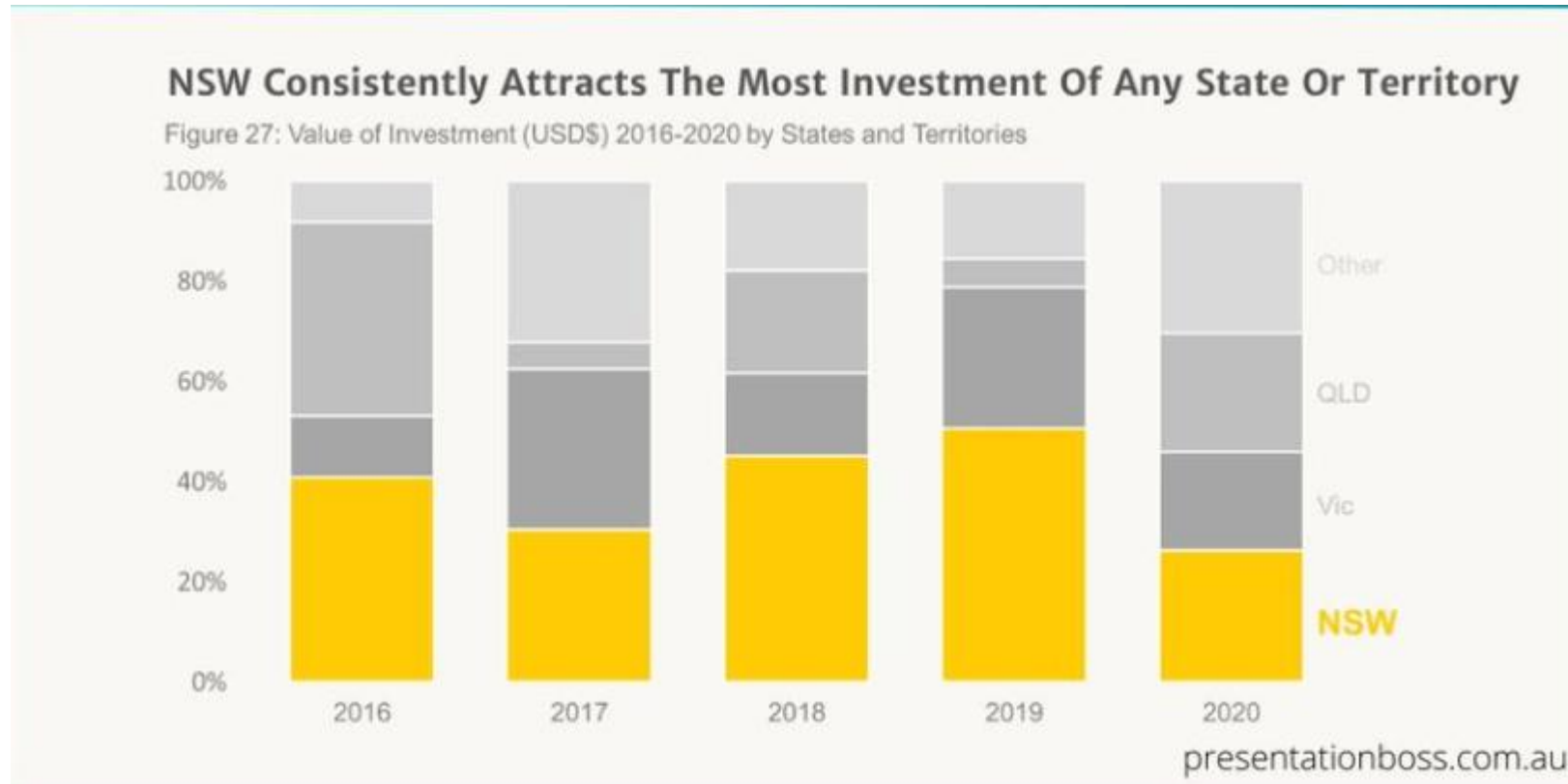
Importance of data visualisation

What is the message?



Importance of data visualisation

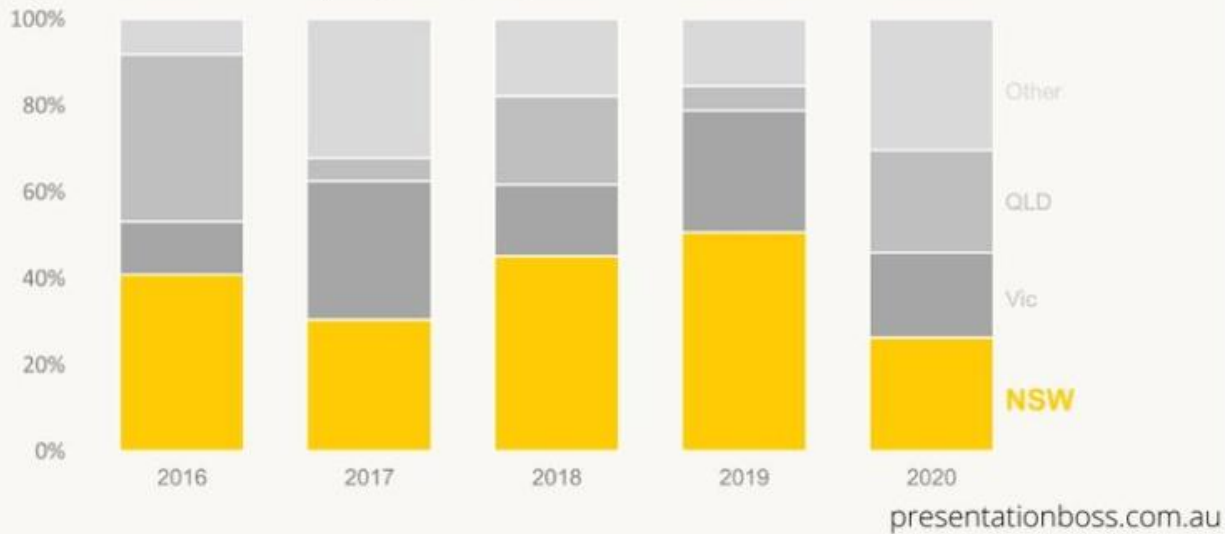
What is the message?



Importance of data visualisation

NSW Consistently Attracts The Most Investment Of Any State Or Territory

Figure 27: Value of Investment (USD\$) 2016-2020 by States and Territories

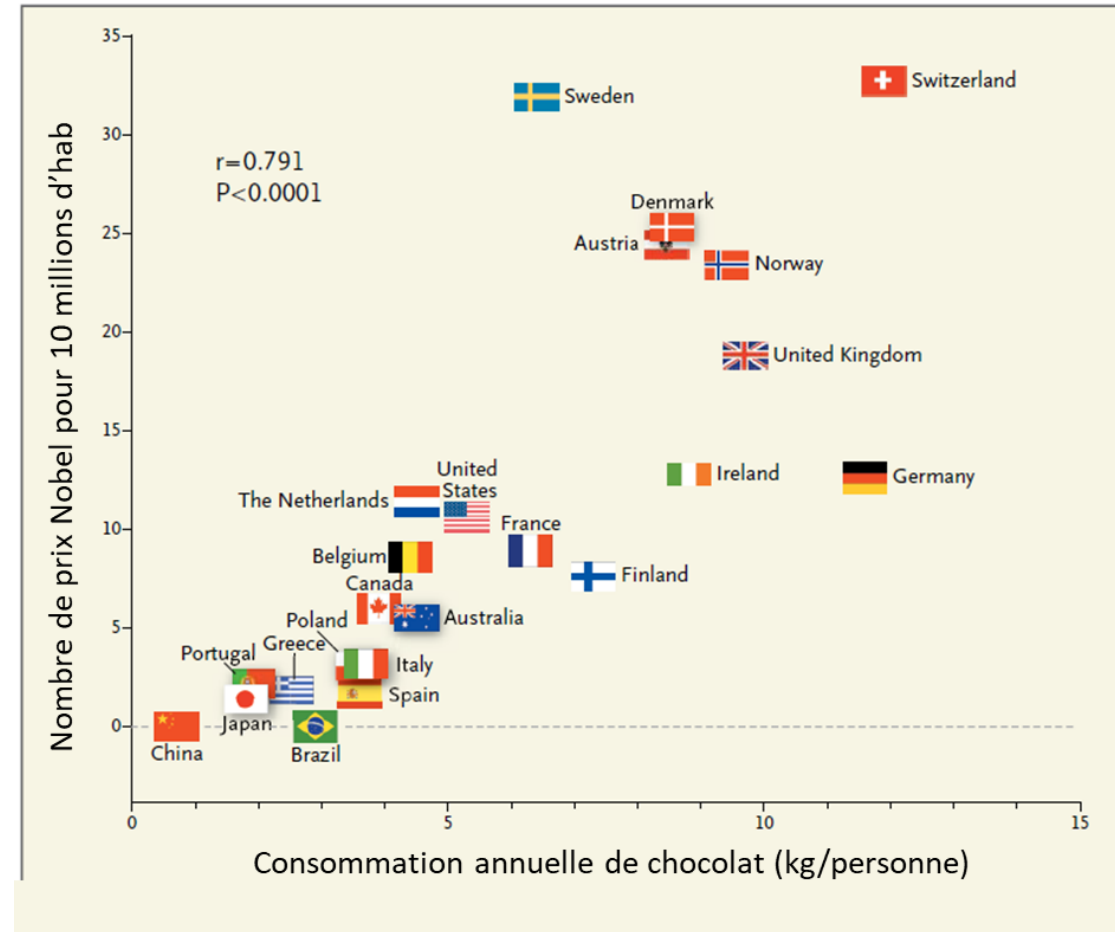


- **Informative** The visualization should be able to convey the desired information from the data to the reader.
- **Efficient** The visualization should not be ambiguous.
- **Appealing** The visualization should be captivating and visually pleasing.

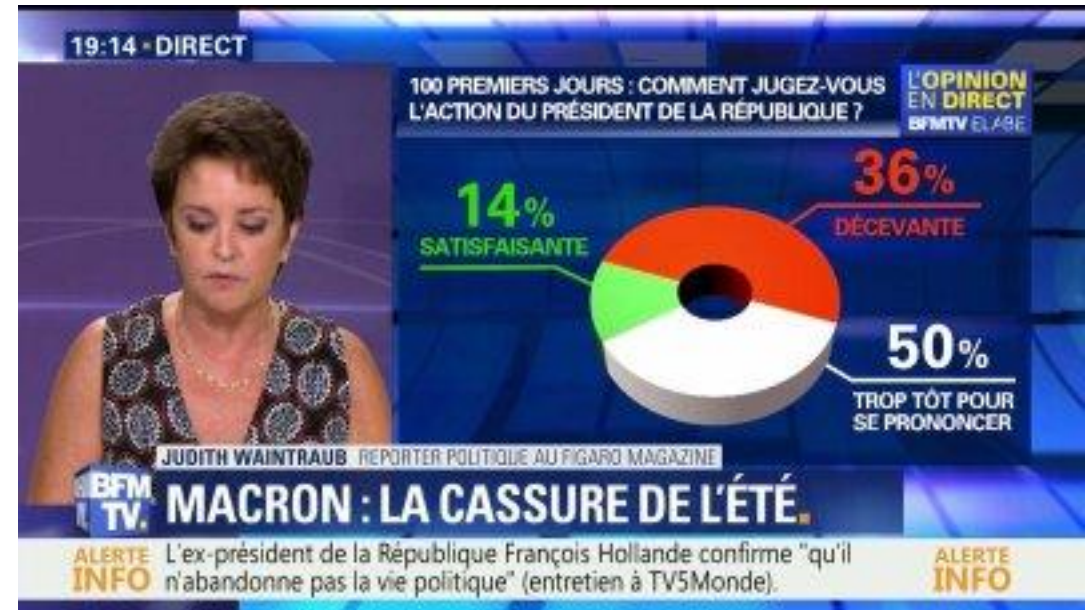
Cons

Visualization has real risks:

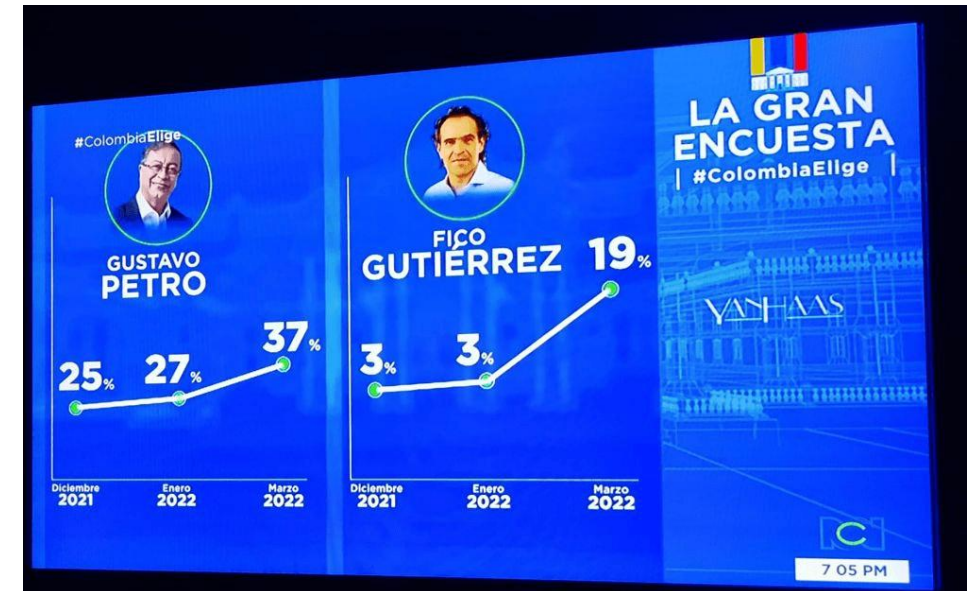
- **Misleading design:** truncated axes, cherry-picked data, 3D effects
- **Oversimplification:** hides uncertainty or nuance
- **Cognitive bias:** viewers may interpret visuals emotionally
- **False authority:** “it looks professional, so it must be true...”
- **Accessibility issues:** poor color choices exclude color-blind users
- **Correlation shown as causation**



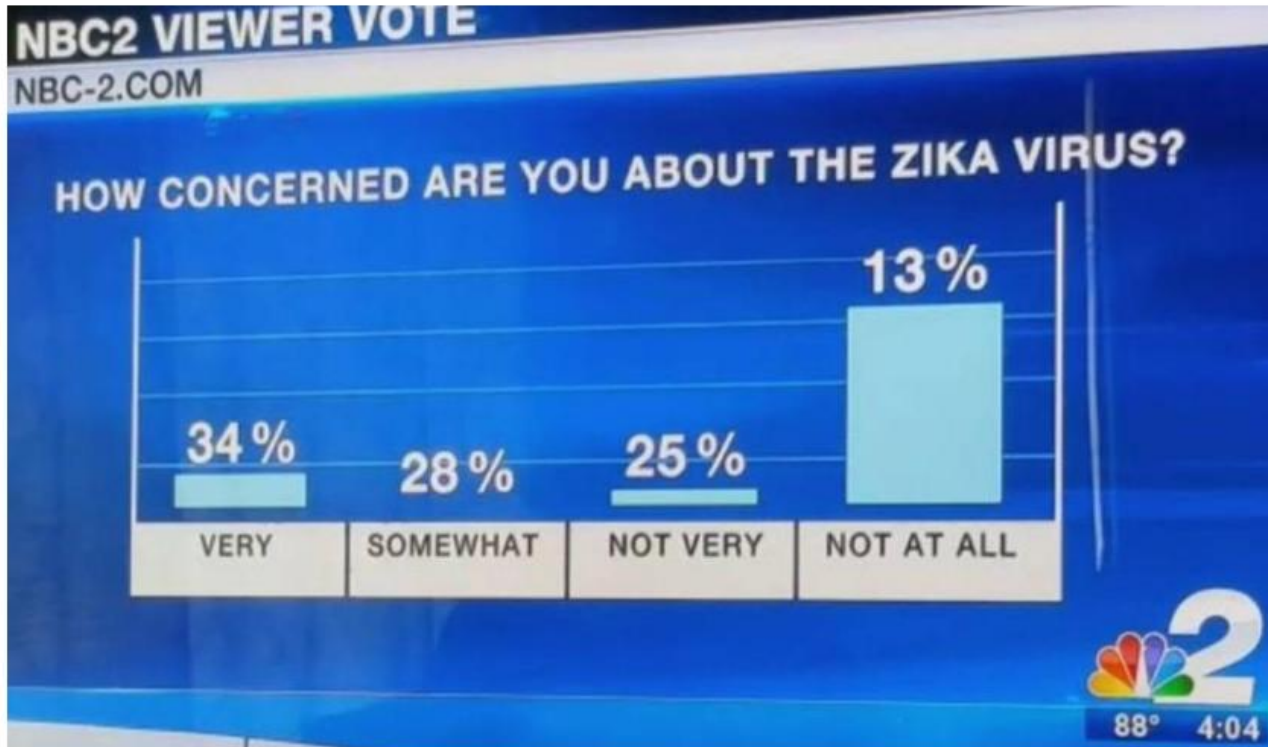
When Visuals Mislead



When Visuals Mislead

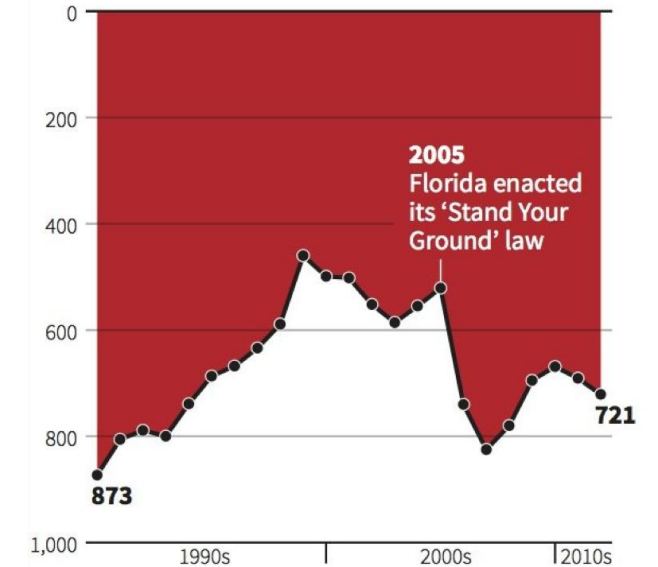


When Visuals Mislead



Gun deaths in Florida

Number of murders committed using firearms



Source: Florida Department of Law Enforcement

C. Chan 16/02/2014

REUTERS

Before You Plot — Structure Your Data

You can't visualize what you haven't cleaned!

What is Data Structuring?

(Also called *data wrangling*, *data preparation*, or *data engineering*)

- Remove duplicates and errors (e.g. misspelling)
- Handle missing values (drop, fill, interpolate)
- Convert formats (e.g. dates, categories)
- Normalize or scale values
- Reshape data (pivot, melt) for plotting

Types of Variables – What Kind of Data Are We Dealing With?

Two Main Types of Variables:

1. Categorical (Qualitative)

- Describes qualities or categories
- Cannot be meaningfully measured or averaged

Examples:

- Gender
- Eye color

Subtypes:

- Nominal: no order (e.g., eye color)
- Ordinal: with order (e.g., satisfaction level)

2. Numerical (Quantitative)

- Represents measurable quantities
- Can be used in mathematical operations

Examples:

- Age
- Height

Subtypes:

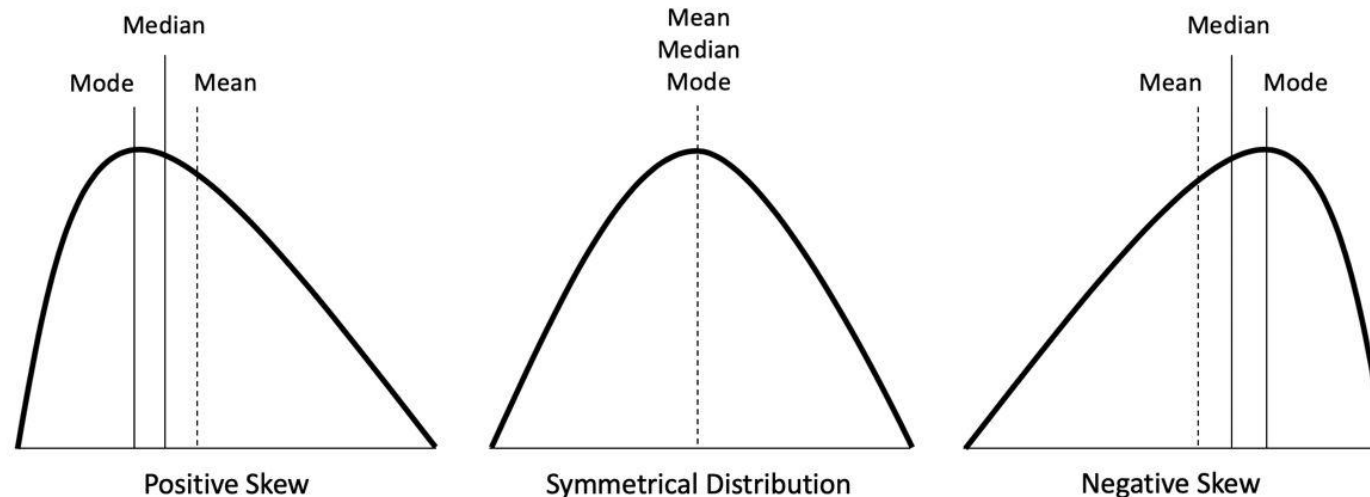
- Discrete: countable (e.g., exams passed)
- Continuous: measurable (e.g., weight, time)

Mean, median mode

Mean: average value computed by adding up all values in a dataset and then dividing the sum by the number of values

Median: The middle value in a set of ordered data.

Mode: The most frequent value in the dataset.



Mean, median mode

You have a list with the following values:

[1, 2, 2, 2, 2, 2, 3, 4, 6, 7, 8, 9, 9]

- What is the mean?
- What is the median?
- What is the mode?

How do these values change if we replace the last 9 with the number 100?

Mean, median mode

You have a list with the following values:

[1, 2, 2, 2, 2, 2, 3, 4, 6, 7, 8, 9, 9]

- What is the mean? **4.38**
- What is the median? **3**
- What is the mode? **2**

How do these values change if we replace the last 9 with the number 100? **Mean changes to 11.38, mode and median the same**

Percentages, Percentiles & Quartiles – What's the Difference?

Percentage (%):

- A way to express a portion of a whole (out of 100)
- Example: 60% of survey respondents chose Option A
- Used in Pie Chart

Percentile

- A value below which a given percentage of data falls
- Example: 90th percentile = better than 90% of the group
- Used in standardized tests, income distribution, etc.

Quartiles

Special percentiles that divide data into four equal parts:

- Q1 (25th percentile) – 25% of values are below this
- Q2 (50th percentile) – the median
- Q3 (75th percentile) – 75% of values are below this
- Used to describe spread in boxplots

Percentages, Percentiles & Quartiles – What's the Difference?

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Ex: weights in the family: 25, 17, 32, 11, 40, 35, 13, 5, and 46.

Percentile

- A value below which a given percentage of data falls
- Example: 90th percentile = better than 90% of the group
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- Calculate the percentages of people with a weights greater than the average
- Calculate the percentages of people with a weights greater than the median

Quartiles

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- Calculate the lower and upper quartiles
- Calculate the 90th percentile

Percentages, Percentiles & Quartiles – What's the Difference?

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Percentile

- A value below which a given percentage of data falls
- Example: 90th percentile = better than 90% of the group
- Used in standardized tests, income distribution, etc.

- Calculate the percentages of people with a weights greater than the average: **55%**

- Calculate the percentages of people with a weights greater than the median: **44%**

Quartiles

Special percentiles that divide data into four equal parts:

- Q1 (25th percentile) – 25% of values are below this
- Q2 (50th percentile) – the median
- Q3 (75th percentile) – 75% of values are below this
- Used to describe spread in boxplots

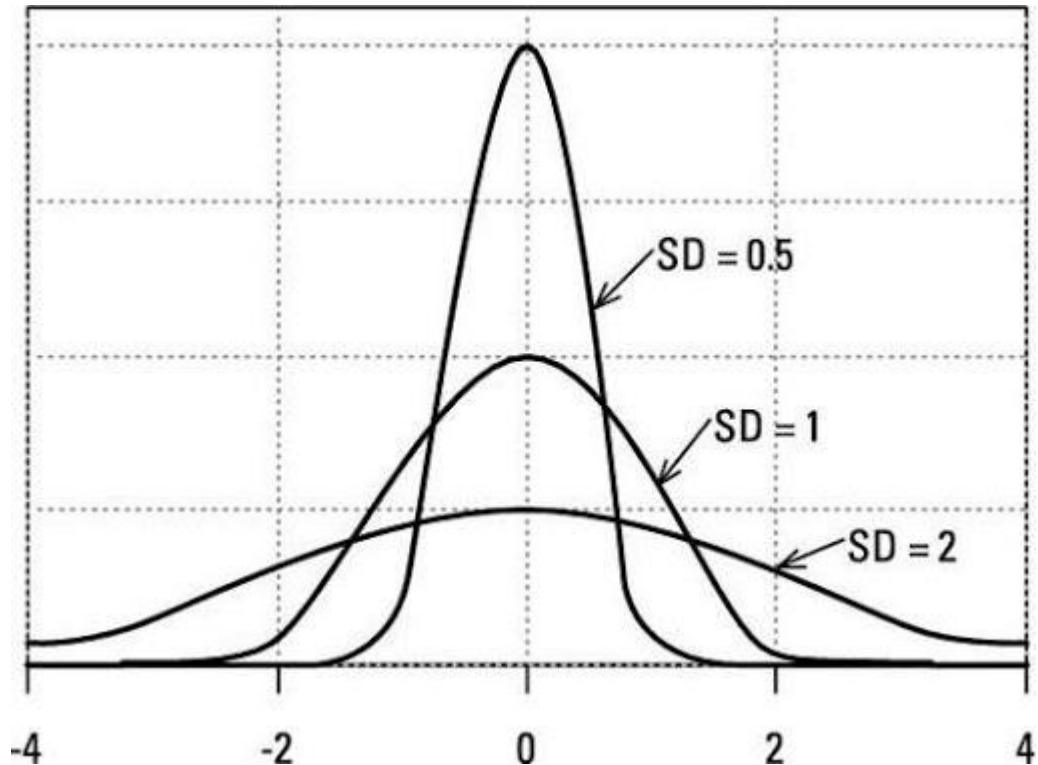
- Calculate the lower and upper quartiles of the following: **12,37.5**

- Calculate the 90th percentile: **$P_k = k(N+1)/100$**
-> **$P_{90} = 90 * 10 / 100 = 9^{th} \rightarrow 46$**

Standard deviation

Standard deviation: The square root, of the mean squared difference of values from the mean, to measure the dispersion of a data set from the mean.

The lower the standard deviation, the more the data is gathered around the mean. The larger the standard deviation, the more dispersed and far from the average the data is.



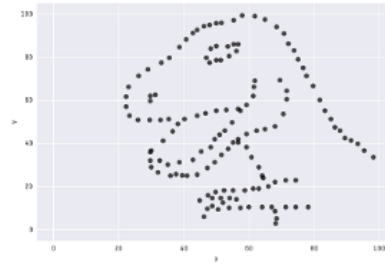
$$\sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

n is the number of persons

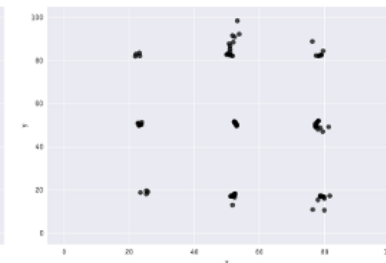
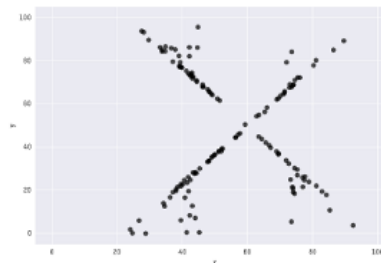
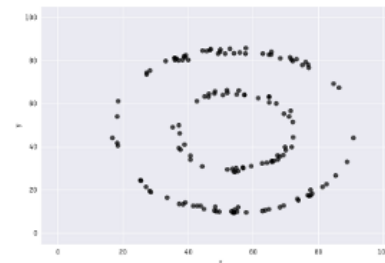
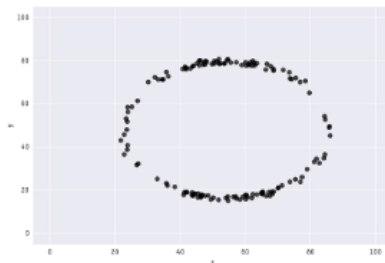
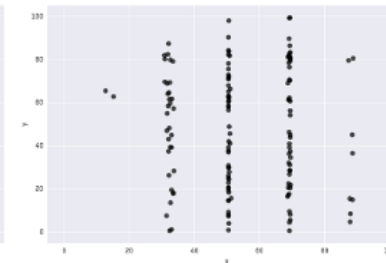
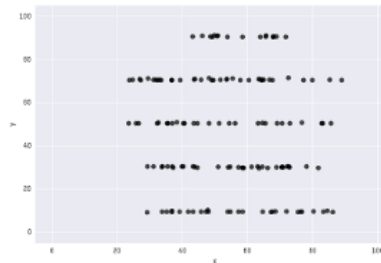
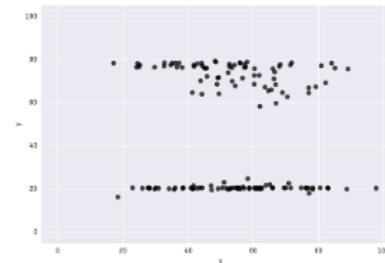
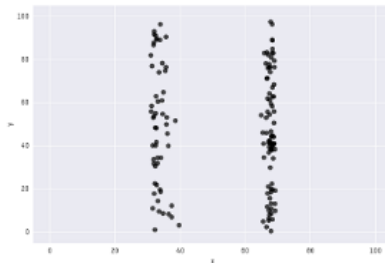
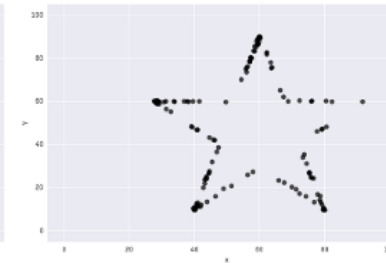
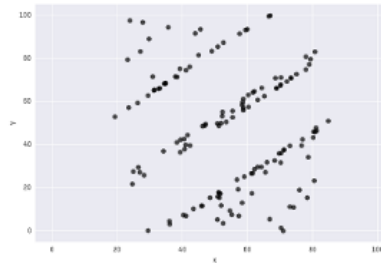
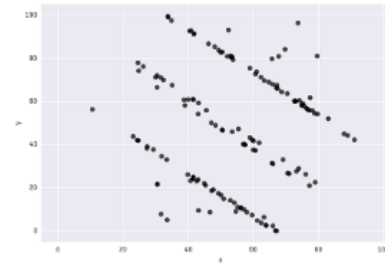
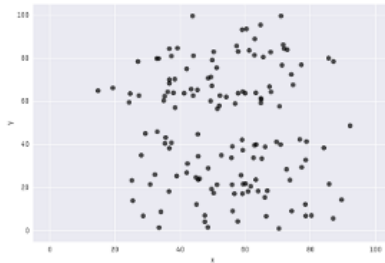
x_i is the size of the individual

$\bar{x}$ is the mean value of all persons

Attention



X Mean: 54.26
Y Mean: 47.83
X SD : 16.76
Y SD : 26.93
Corr. : -0.06



Probabilities

Probability is the measure of the possibility of an event occurring in a random experiment.

Probability is a number between **0 and 1**
0 = impossible, 1 = certain

The probability of an event can therefore be measured as the number of favorable outcomes divided by the total number of possible outcomes.

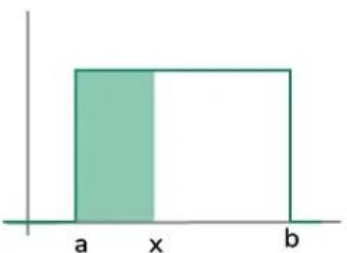
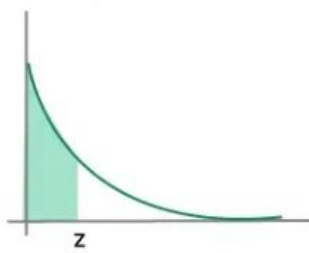
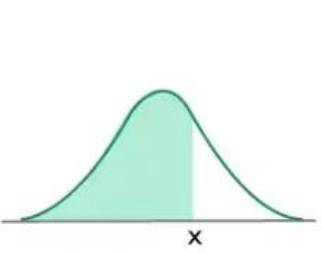
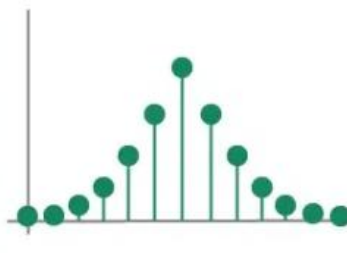
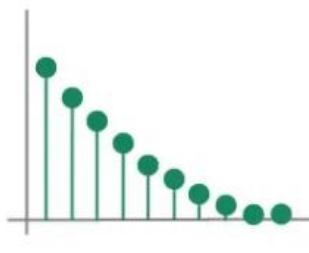
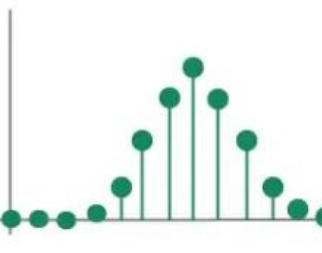
Event	Probability
Flipping a coin – Heads	0.5
Rolling a 6 on a die	$1/6 \approx 0.167$
Drawing an ace from a deck	$4/52 \approx 0.077$

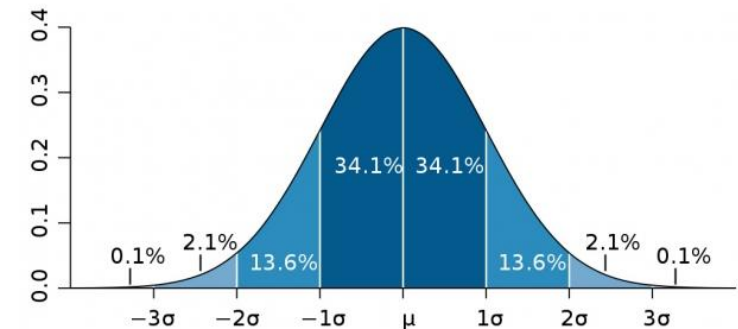
Distributions

The distribution of observed probabilities can take several forms. It provides a way of **modeling the likelihood of each outcome in a random experiment**.

Probability Distribution



Continuous	Uniform  A graph of a uniform distribution showing a constant probability density between values a and b . A point x is marked within this interval.	Exponential  A graph of an exponential distribution showing a probability density that decays as the value increases. A point z is marked on the x-axis.	Normal  A graph of a normal distribution showing a symmetric bell curve. A point x is marked on the x-axis.
	Binomial  A graph of a binomial distribution showing discrete outcomes with a symmetric, bell-shaped distribution of bars.	Geometric  A graph of a geometric distribution showing discrete outcomes with a probability mass function that decays exponentially.	Hypergeometric  A graph of a hypergeometric distribution showing discrete outcomes with a symmetric, bell-shaped distribution of bars.



68% of values fall within $\pm 1\sigma$ of the mean

~95% within $\pm 2\sigma$

~99.7% within $\pm 3\sigma$

Population vs Sample – Who Are We Studying?

Definition

In statistics, a population is the set on which a statistical study is conducted. A sample is a (usually known) subset of the population

Population

→ The **entire group** we are interested in studying

Examples: all French voters, all EPITA students, all galaxies in the observable universe

Sample

→ A **subset** of the population that we actually collect data from

Examples: 1,000 survey respondents, 50 students, 10,000 galaxies from SDSS

Why use a sample?

- **Faster**
- **Cheaper**
- Often **more practical or ethical**

Goal: Make **valid conclusions** about the population **from** the sample

Project: where to get data?

Explore the following websites to find one dataset that you would like to work on:

- <https://data.europa.eu/en>
- <https://data.worldbank.org/>
- <http://data.un.org/>
- <https://datasetsearch.research.google.com/>
- <https://github.com/datasets>
- <https://opendata.paris.fr/pages/home/>
- <https://www.data.gouv.fr/fr/>

You can also identify a topic you would be interested to work on

Project: what is a licence?

“Licences tell you **what you can do** with the content or data that you access.

A licence will tell you whether you can:

- republish the content or data on your own website
- derive new content or data from it
- make money by selling products that use it
- republish it while charging a fee for access

If you break the terms of the licence, the owner of the content or data can take you to court.”

https://data.europa.eu/sites/default/files/d2.1.2_training_module_2.5_data_and_metadata_licensing_en_edp.pdf

Project: type of licence

CREATIVE COMMONS LICENSES		 COPY & PUBLISH	 ATTRIBUTION REQUIRED	 COMMERCIAL USE	 MODIFY & ADAPT	 CHANGE LICENSE								
	PUBLIC DOMAIN	✓	✗	✓	✓	✓								
	CC BY	✓	✓	✓	✓	✓								
	CC BY-SA	✓	✓	✓	✓	✗								
	CC BY-ND	✓	✓	✓	✗	✗								
	CC BY-NC	✓	✓	✗	✓	✓								
	CC BY-NC-SA	✓	✓	✗	✓	✗								
	CC BY-NC-ND		✓	✗	✗	✗								
	You can redistribute (copy, publish, display, communicate, etc.)			You have to attribute the original work			You can use the work commercially			You can modify and adapt the original work			You can choose license type for your adaptations of the work.	
The Creative Commons licences explained.														

More infos at:
<https://creativecommons.org/>

Project: General Data Protection Regulation

All collected, stored and processed data of european citizen need to adhere to the GDPR.

Key aspects of the GDPR are:

- **Data Minimization:** Collect only the data necessary for the intended purpose.
- **Lawful Processing:** Data processing must have a legal basis (consent, contract, legal obligation, vital interests, public task, legitimate interests).
- **Transparency:** Inform individuals about data processing activities.
- **Data Security:** Implement measures to protect data from breaches. Encourage privacy by design.
- **Individual Rights:** Individuals have the right to access, rectify, erase, and object to the processing of their data.

Non-compliance can result in significant fines, with fines depending on the severity of the violation.

Project: what to do after gathering data

One of the most crucial aspect in data analysis is understanding the background of your data. Without this knowledge, it's challenging to make informed decisions, detect potential biases, or effectively communicate findings, hindering the value and trustworthiness of data-driven analyses.

- What data was collected ? What is the data type of each column ?
- When was the data collected ?
- Who collected the data ? Who financed the collection of the data
- For which purpose was the data collected ?
- What is the licence of the data set ?
- How much data was collected ?
- Are there any missing data ? Are there any duplicates ?

Project: Research question

Every project should have a research question. It is a question that you are curious about and that you want to answer. Simply put, a research question gives you a North star, a sense of direction

It should be:

- Focused on a **single problem or issue**
- **More about explanation than description**
- **Researchable:** using primary and/or secondary sources
- **Feasible** to answer within the timeframe (project deadline is in a few weeks)
- Specific enough to answer thoroughly
- Complex enough to develop the answer: a simple Google search should not answer the question!!

Example: Subject: Men's Long Jump World Records

- What does the progression of world records look like over time?
- Is the progression steady, or are improvements linked to external factors like better training, nutrition, or technology?
- How does the men's long jump world record progression compare to the women's progression?

Project: Research question

Useful Guiding Questions

1. What are the main features of X?
2. In what ways has X evolved over time?
3. What factors contribute to the existence of X?
4. How does X connect with Y?
5. What effects does X have on Y?
6. What are the strengths and weaknesses of X?
7. What steps could make X better?

AVOID:

- **not be based on value judgements** (Is X better than Y?)

- **vague language, jargon, and too-broad ideas.**

Bad: What effect does social media have on people's minds?

Good: What effect does daily use of tiktok have on the attention span of 16-year-olds at high school?

- **not be answered with yes or no**

Project: Research question

Once you've defined your research question, the next step is to organize your analysis.

In most cases, a single overarching question is too broad to tackle all at once. Breaking it down into smaller, focused subquestions allows you to approach the problem methodically and build up to a comprehensive answer step by step.

Helpful Tips

Clear sub-questions should be:

1. Easier to address than the main question
2. Focused on a single aspect
3. Organized in a sensible sequence

Project: Research question

YOUR TURN

- **Select a dataset**
- **Explore the dataset**
- **Write a research question**
- **Define sub questions**